Ethan Villalovoz

(530) 558-1523 | ethan.villalovoz@gmail.com | LinkedIn | GitHub | ethanvillalovoz.com

Education

Georgia Institute of Technology — **Expected May 2028**
M.S. in Computer Science (Computational Perception & Robotics) | GPA: 4.0/4.0 — Atlanta, GA

Washington State University, Honors College — **Aug 2021 – May 2025**
B.S. in Computer Science, Minor in Mathematics | GPA: 3.94/4.0, Summa Cum Laude — Pullman, WA

- **Honors & Awards:** CS Research Mentorship Program Scholar, Google Research; Generation Google Scholarship

Experience

Georgia Institute of Technology — **Feb 2026 – Present**
Graduate Student Researcher — Atlanta, GA

- Adapting a **Wan2.2-TI2V-5B video-to-action codebase** to train and evaluate robot policies on **LIBERO** by adding dataset conversion, action normalization, and PyTorch/Hydra pipelines.

Microsoft — **May 2026 – Jul 2026**
Software Engineer Intern — Redmond, WA

- Built a C#/.NET 10 proof-of-concept **AI agent for financial close** using Azure AI Foundry, Azure OpenAI, and GraphQL APIs, giving users one natural-language interface to investigate blockers across multiple applications.
- Cut measured AI inference cost about **30×** while matching GPT-5.4 tool-routing quality by benchmarking 7 Azure OpenAI models, selecting GPT-5.4-nano, curating its tools, and correcting routing failures with targeted instructions.
- Increased agent throughput about **22× (0.5 to 11 requests/second)** by load-testing concurrency, tracing throttling to undersized Azure OpenAI capacity, and resizing the deployment with no observed throttling failures.
- Resolved **22 security findings** across two production repositories, including preventing unauthorized file access across four C# endpoints through layered path and permission validation backed by 67 tests.

Carnegie Mellon University — **Jun 2024 – Aug 2024**
Robotics Institute Summer Scholar — Pittsburgh, PA

- Developed a **Python Bayesian reward-learning simulation** for household robotics, modeling user preferences with hierarchical decision trees over object type, color, material, and placement.
- Implemented an **interactive MDP** that updated beliefs from simulated object-placement corrections and follow-up feature questions, ranking the ground-truth preference model highest in a rollout.

Google — **May 2023 – Aug 2023**
Software Engineering Intern (STEP) — Sunnyvale, CA

- Developed and deployed **5 C++/SQL statistics-collection jobs** for pending queues in an internal database, replacing manual analysis with automated metric reporting.
- Scaled statistics collection from **1% to 100% of the database in under 4 hours** by implementing incremental job sampling, reducing anticipated runtime by 66%.

Projects

Knowledge Graph RAG Assistant | React, TypeScript, FastAPI, FAISS, DBpedia/SPARQL, OpenAI, Docker

- Generated and indexed SentenceTransformer embeddings from **5GB+ of Wikipedia data** spanning **10,000+ articles** for a RAG assistant that answers questions using FAISS semantic search and DBpedia knowledge-graph context.

Publications

An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models. S. Tomar, A. Deshwal, **E. Villalovoz**, M. Fazzini, H. Cai, J. R. Doppa. *arXiv:2512.15076*, 2025.

Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach. A. Bacula, **E. Villalovoz**, D. Flynn, A. Mehta, H. Knight. *IROS*, 2023.

Technical Skills

Languages & Scripting: Python, C/C++, C#, SQL, TypeScript, JavaScript, Bash, PowerShell
AI/ML & Robotics: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, FAISS, OpenCV, ROS
Web/Backend: .NET, ASP.NET Core, GraphQL, React, Next.js, FastAPI, Flask, REST APIs
Cloud & Tools: Microsoft Azure, Azure AI Foundry, Azure OpenAI, AWS, Docker, Git, GitHub Actions